

eDNA Cost-Benefit Analysis: Optimizing COTS Control

AUTHOR

Analysis Report

PUBLISHED

April 24, 2026

► Code

Executive Summary

This report evaluates the potential return on investment (ROI) of incorporating eDNA surveys into the Crown-of-Thorns Starfish (COTS) control program to safely identify and “close” tactical reefs. By eliminating “ghost culling”—the effort spent culling reefs that lack significant COTS populations—the program could redirect significant vessel and diver time.

This analysis relies on actual historical cull efficiency data matched with eDNA records since 2020.

Baseline Unit Costs

- **Cost per Diver Hour:** \$1,000
 - **eDNA Program Total Cost:** \$300,000
 - **Assumed Number of Tactical Reefs (per year):** 40 (Assumes 20 targeted reefs + repurposing 20 monitoring reefs)
 - **eDNA Average Cost per Reef:** \$2,500
-

Historical eDNA Efficacy Assessment (2020-Present)

Rather than just looking at the worst performing reefs, we have broadened our search to **all reefs that have both eDNA and active culling data since 2020.**

We evaluate eDNA rigorously as a diagnostic test to see what would have happened if we explicitly closed reefs based on statistical combinations.

Classification Logic & Defining “Ghost Culls”

To rigorously score False Positives and False Negatives, we must define the ultimate ground truth. * **The Predictor Rule (Prediction):** A statistical cutoff indicating whether we should “Close” our vessel operations or keep the reef “Open” for diving. * **The Ground Truth (Actual Cull Outcome):** * *Current Assumption:* An **Outbreak (Needed)** is formally declared if *any* dive on that reef registered a CPUE > 0.04. If a reef met this criteria, diving was validated. * *What is a Ghost Cull?* A **Ghost Cull (False Positive)** is classified when the predictor rule says we should keep the reef open to cull it, but following the initial Manta Tows/scouts, the subsequent Culling Data proves that **zero dives ever exceeded 0.04 CPUE**. In other words, if a reef actually had dives recovering > 0.04 CPUE, it is completely exonerated—it is *not* a ghost cull, and our models strictly punish the metric (registering a False Negative Missed Outbreak) if we mistakenly attempted to close it.

Temporal Correction: Post-eDNA Filtering

To avoid inflating the estimated savings, **all culling metrics are computed using only dives that occurred on or after the earliest eDNA sample date** for each reef-year. This ensures:

1. **Hours reported as “avoidable” could actually have been avoided** — we only count culling effort that happened after the eDNA result would have been available to inform the closure decision.
2. **Ghost-cull classification is not contaminated by pre-eDNA evidence** — a reef is not falsely exonerated (or falsely condemned) based on CPUE data that preceded the eDNA sample.

For reef-years where all culling occurred before the eDNA sample, those records are excluded entirely from the confusion matrix (there is no post-eDNA culling to evaluate).

Baseline Confusion Matrix (eDNA Only)

If we applied the $< 45\%$ rule retrospectively to all fully matched reef/years, these are the outcomes:

► Code

Baseline eDNA Confusion Matrix (with Diver Hour bins)

Confusion_Class	Reef Occurrences	Diver Hrs Rep	0-0.02 Hrs (%)	0.02-0.04 Hrs (%)	0.04-0.08 Hrs (%)	>0.08 Hrs (%)
True Positive (Correctly Kept Open)	39	8371.5	27.7%	19.4%	25%	27.9%
True Negative (Correctly Closed)	18	859.8	93.9%	6.1%	0%	0%
False Positive (Ghost Culled)	12	512.8	89.2%	10.8%	0%	0%
False Negative (Missed Outbreak)	4	583.2	52.1%	21.7%	20.4%	5.8%

Multivariate Diagnostic Engine

To aggressively eliminate False Negatives (missed outbreaks) and False Positives (wasted culling trips), we can synthesize additional site data. By bringing in **Manta Tow Scars** and applying an **Initial Scout Cull** (recording CPUE strictly from the first day on-site before committing heavily to the reef), we enhance prediction accuracy.

Baseline Comparison: Manta Scars Only

To fully understand the value eDNA adds, we must evaluate what happens if we operate solely on traditional Manta Tow scarring data. **Close Reef IF Manta Scars $< 10\%$:**

► Code

Univariate Baseline: Scars $< 10\%$ (No eDNA factors)

Confusion_Class_Scars	Reef Occurrences	Diver Hrs Rep
True Positive (Correctly Kept Open)	30	8126.2
True Negative (Correctly Closed)	22	935.6
False Negative (Missed Outbreak)	13	828.4
False Positive (Ghost Culled)	8	437.0

Hardcoded Rule: eDNA combined with Manta Scars

If we refine the rule to mathematically state: **Close Reef IF (eDNA < 45%) AND (Manta Scars < 10%):**

► Code

Multivariate Rule 1 (eDNA + Scars)

Confusion_Class_R1	Reef Occurrences	Diver Hrs Rep
True Positive (Correctly Kept Open)	40	8796.2
False Positive (Ghost Culled)	16	647.9
True Negative (Correctly Closed)	14	724.7
False Negative (Missed Outbreak)	3	158.4

Baseline Comparison 2: Field Data Only (Scars + Scout Cull)

Before utilizing expensive eDNA processing, what happens if we strictly merge our free, existing field data?

Close Reef IF (Manta Scars < 15%) AND (Initial Scout CPUE < 0.02). *(Note: Because this rule inherently requires the Initial Scout Cull, we deduct Scout Diver Hours from the True Negative yield).*

► Code

Baseline 2: Field Rules strictly (Scars < 15% + Scout Culling < 0.02)

Confusion_Class_Field	Reef Occurrences	Net Diver Hrs Rep
True Positive (Correctly Kept Open)	35	8384.0
True Negative (Correctly Closed)	23	763.0
False Negative (Missed Outbreak)	8	570.6
False Positive (Ghost Culled)	7	240.3

Hardcoded Rule 2: eDNA, Manta Scars, and Initial Scout CPUE

If we deploy a highly integrated diagnostic expanding the relaxed envelopes but demanding an in-water check:

Close Reef IF (eDNA < 50%) AND (Manta Scars < 15%) AND (Initial Scout CPUE < 0.02): *(Note: Because this rule requires the Initial Scout Cull, we mathematically deduct the Scout Diver Hours from the functionally Saved Hours yield).*

► Code

Multivariate Rule 2 (eDNA + Scars + Scout Culling)

Confusion_Class_R2	Reef Occurrences	Net Diver Hrs Rep
True Positive (Correctly Kept Open)	41	8816.3
True Negative (Correctly Closed)	18	490.3
False Positive (Ghost Culled)	12	599.1
False Negative (Missed Outbreak)	2	138.4

Algorithmic Optimization (Decision Trees via rpart)

A decision tree ([rpart](#)) mathematically scans historical records across eDNA, Manta Tow, and Initial CPUE variables to determine absolute optimal thresholds for closure decisions. *(Note: Because these models inherently require divers doing an initial scout cull to get [Initial_CPUE](#) , the ultimate “Saved Hours” in these models formally subtracts those day-1 Scout Diver Hours from the savings yield).*

Instead of blindly eliminating all False Positives or False Negatives completely, we present three models based on standard statistical weighting metrics:

► [Code](#)

Model 1: Balanced Accuracy

Focus: Treats Ghost Culling and Missed Outbreaks equally (1:1 ratio). Optimizes purely for the highest raw predictive accuracy (Gini index). This represents an operational baseline.

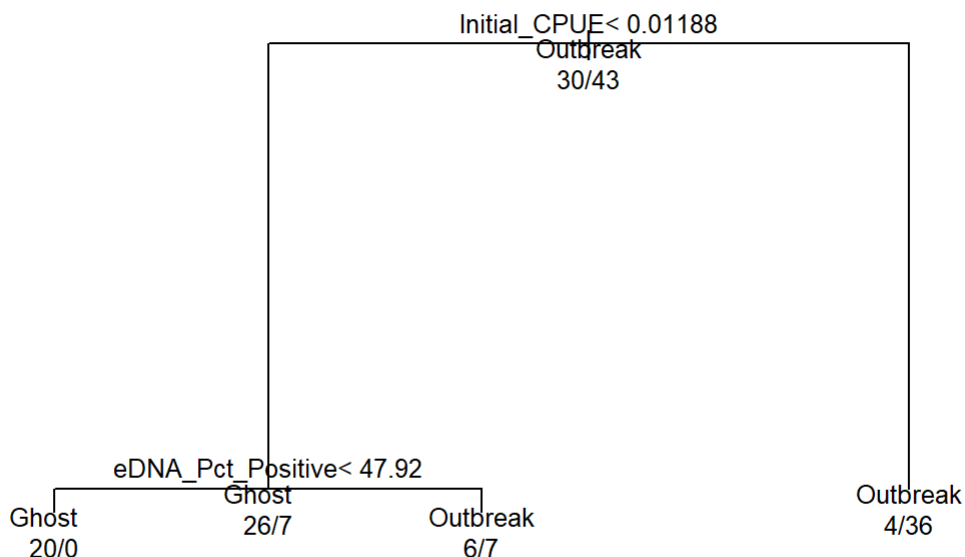
► [Code](#)

Model 1: Balanced Accuracy Outcomes

Class_Bal	Reef Occurrences	Net Diver Hrs
False Positive (Ghost Culled)	10	562.2
True Negative (Correctly Closed)	20	522.6
True Positive (Correctly Kept Open)	43	8954.7

► [Code](#)

Threshold Splitting Rules: Balanced Accuracy



Model 2: F1-Score Proxy (Harmonic Mean)

Focus: The F1 Score acts as the harmonic mean between Precision and Recall. Because there are fewer real outbreaks than ghost reefs (class imbalance), we apply a calculated 2.5-to-1 penalty for missing an outbreak to achieve mathematically resilient F1 classification.

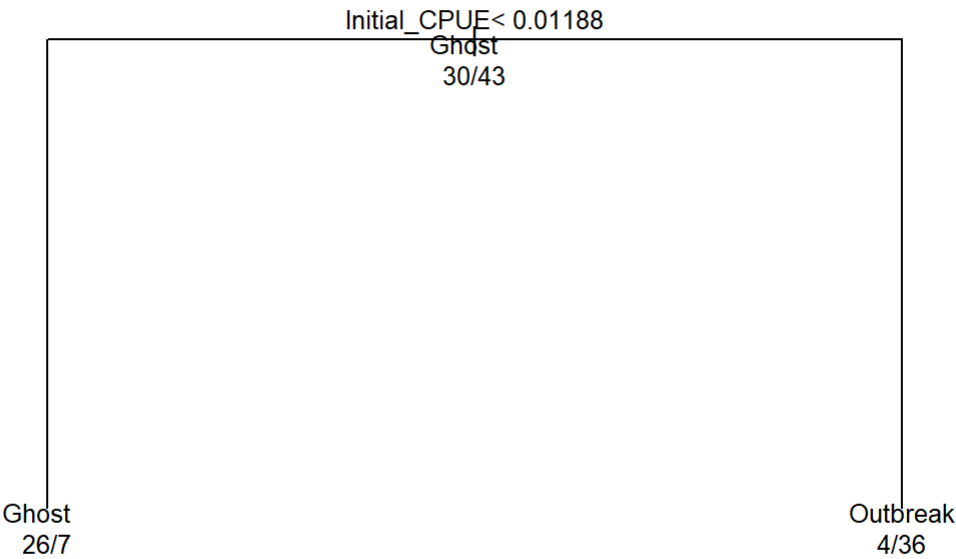
► Code

Model 2: F1-Score Harmonic Mean Outcomes

Class_F1	Reef Occurrences	Net Diver Hrs
False Negative (Missed Outbreak)	7	1597.0
False Positive (Ghost Culled)	4	151.9
True Negative (Correctly Closed)	26	821.5
True Positive (Correctly Kept Open)	36	7357.6

► Code

Threshold Splitting Rules: F1-Score



Model 3: F2-Score Proxy (Eco-Conservative)

Focus: The F2 Score weights Recall (Outbreak Detection) twice as heavily as Precision. We reflect this by imposing a 5-to-1 penalty against False Negatives. This guarantees an ecologically safe approach without collapsing into absolute zero-risk rigidity.

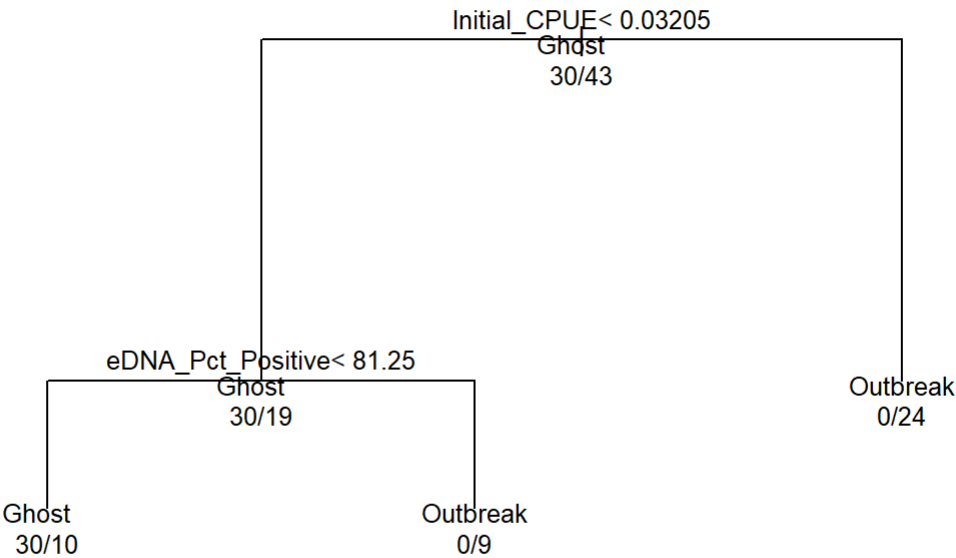
► Code

Model 3: F2-Score Eco-Conservative Outcomes

Class_F2	Reef Occurrences	Net Diver Hrs
False Negative (Missed Outbreak)	10	1090.8
True Negative (Correctly Closed)	30	904.5
True Positive (Correctly Kept Open)	33	7863.9

► Code

Threshold Splitting Rules: F2-Score



Potential Scale vs Operational Program

The confusion matrices above are based on our historical validation dataset — a relatively small pool of reef-years where we have **both** eDNA samples and post-eDNA culling data. To estimate the annual value of deploying these rules at operational scale, we extrapolate from the historical per-reef averages to a hypothetical program size.

Step 1: Key Assumptions

► Code

Scaling Assumptions

Parameter	Value
Cost per on-water day	\$24,000

Parameter	Value
Diver hours per on-water day	24 hrs
Implied cost per diver hour	\$1,000
eDNA program total annual cost	\$300,000
Total reefs surveyed with eDNA	120
eDNA cost per reef (program_cost / reefs_surveyed)	\$2,500
Base tactical reefs (actively targeted for closure)	20
Monitoring reefs repurposed for tactical use	20
Total operational tactical reefs per year	40

Step 2: Historical Per-Reef Averages (from validation data)

We compute the **average hours saved** and **average missed outbreaks** per reef-year across the entire historical validation dataset, then scale those averages to the operational program size.

► Code

Number of Unique Historical Target Validations per Year

Year	Reefs Tested
2020	6
2021	13
2022	12
2023	17
2024	4
2025	21

► Code

Step 2: Historical Per-Reef Averages

Metric	Approach A: Field Only	Approach B: Integrated eDNA
Total reef-years in validation dataset	73	73
True Negatives (reefs correctly closed)	23	18
Total post-eDNA diver hours saved (TN reefs, net of scout hrs)	763 hrs	490.3 hrs
False Negatives (missed outbreaks)	8	2
Avg hours saved per reef-year (TN_hrs / total_reefs)	10.45 hrs/reef	6.72 hrs/reef
Avg missed outbreaks per reef-year (FN_count / total_reefs)	0.11	0.027

Step 3: Scale to Operational Program

We multiply the per-reef averages by the operational program size (40 reefs/year) and convert hours to dollars.

► Code

Step 3: Scaled to Operational Program

Metric	Approach A: Field Only	Approach B: Integrated eDNA
Avg hrs saved/reef × 40 reefs	10.45 × 40	6.72 × 40
Projected annual hours saved	418.1 hrs	268.7 hrs
× \$1,000/hr = Gross dollar savings	\$418,064	\$268,658
Avg FN/reef × 40 reefs	0.11 × 40	0.027 × 40
Projected missed outbreaks per year	4.4	1.1
eDNA testing cost (Approach B only)	\$0 (no eDNA needed)	\$100,000
NET savings after eDNA costs (Approach B only)	\$418,064	\$168,658

Summary

Approach A: Field Data Only (Scars < 15% + Scout CPUE < 0.02)

- **Operating Cost:** \$0 extra equipment.
- **Projected Annual Hours Saved:** 418.1 diver hours.
- **Gross Dollar Savings:** \$418,064.
- **Operational Risk:** Expected to miss **4.4** genuine outbreaks per year.

Approach B: Integrated eDNA (eDNA < 50% + Scars < 15% + Scout CPUE < 0.02)

- **Operating Cost:** \$100,000 for eDNA testing across 40 reefs.
- **Projected Annual Hours Saved:** 268.7 diver hours.
- **Gross Dollar Savings:** \$268,658.
- **NET Savings (after eDNA costs):** \$168,658.
- **Operational Risk:** Expected to miss **1.1** genuine outbreaks per year.

Potential Scale: Manta-Only vs eDNA + Manta (No Scout Culling Required)

The comparison above requires an initial scout cull before a closure decision can be made. Below, we present the simpler rules that only require a Manta Tow (and optionally eDNA), with **no initial cull day needed**. Because no scout culling is involved, the full **Total_Diver_Hours** at the reef is the potential saving (no scout hours deducted).

► Code

Historical Per-Reef Averages: Manta-Only vs eDNA + Manta

Metric	Approach C: Manta Only	Approach D: eDNA + Manta
Total reef-years in validation dataset	73	73
True Negatives (reefs correctly closed)	22	14
Total post-eDNA diver hours saved (TN reefs, no scout deduction)	935.6 hrs	724.7 hrs
False Negatives (missed outbreaks)	13	3

Metric	Approach C: Manta Only	Approach D: eDNA + Manta
Avg hours saved per reef-year (TN_hrs / total_reefs)	12.82 hrs/reef	9.93 hrs/reef
Avg missed outbreaks per reef-year (FN_count / total_reefs)	0.178	0.041

► Code

Scaled to Operational Program: Manta-Only vs eDNA + Manta

Metric	Approach C: Manta Only	Approach D: eDNA + Manta
Avg hrs saved/reef × 40 reefs	12.82 × 40	9.93 × 40
Projected annual hours saved	512.7 hrs	397.1 hrs
× \$1,000/hr = Gross dollar savings	\$512,676	\$397,096
Avg FN/reef × 40 reefs	0.178 × 40	0.041 × 40
Projected missed outbreaks per year	7.1	1.6
eDNA testing cost	\$0 (no eDNA needed)	\$100,000
NET savings after eDNA costs	\$512,676	\$297,096

Summary (No Scout Culling)

Approach C: Manta Scars Only (Scars < 10%)

- **Operating Cost:** \$0 extra.
- **Projected Annual Hours Saved:** 512.7 diver hours.
- **Gross Dollar Savings:** \$512,676.
- **Operational Risk:** Expected to miss **7.1** genuine outbreaks per year.

Approach D: eDNA + Manta (eDNA < 45% + Scars < 10%)

- **Operating Cost:** \$100,000 for eDNA testing.
- **Projected Annual Hours Saved:** 397.1 diver hours.
- **Gross Dollar Savings:** \$397,096.
- **NET Savings (after eDNA costs):** \$297,096.
- **Operational Risk:** Expected to miss **1.6** genuine outbreaks per year.

Deep Dive: Most Recent Example Year

The table below provides transparency out of the most recent example year to show you the hour breakdown by precise density bins alongside their associated metrics enabling multivariate closures.

► Code

Worst-Performing Tactical Reefs in 2025 (Deep Dive Example)

Reef Name	Tot Dive Hrs	Wasted Dive Hrs	% 0-0.02	% 0.02-0.04	% 0.04-0.08	% >0.08	eDNA % Pos	eDNA Mean Conc
-----------	--------------	-----------------	----------	-------------	-------------	---------	------------	----------------

Reef Name	Tot Dive Hrs	Wasted Dive Hrs	% 0-0.02	% 0.02-0.04	% 0.04-0.08	% >0.08	eDNA % Pos	eDNA Mean Conc
Lizard Island Reef (North West) (14-116a)	71.7	38.3	53.5%	46.5%	0%	0%	15%	1.725
U/N Reef (21-057)	28.5	19.0	66.7%	33.3%	0%	0%	16.7%	1.568
Green Island Reef (16-049)	56.7	20.0	35.3%	44.1%	10.3%	10.3%	22.2%	3.201
Tongue Reef (16-026)	29.8	0.0	0%	0%	32.8%	67.2%	38.9%	4.741
U/N Reef (21-069)	11.8	11.8	100%	0%	0%	0%	4.2%	0.348
Lizard Island Reef (North East) (14-116b)	11.2	11.2	100%	0%	0%	0%	45%	7.674
(Big) Broadhurst Reef (No 1) (18-100a)	496.2	261.0	52.6%	17.5%	14.1%	15.8%	50%	7.625
U/N Reef (21-067)	14.3	0.0	0%	0%	25.2%	74.8%	50%	18.442
U/N Reef (20-386)	69.4	18.0	25.9%	18.5%	20.9%	34.7%	52.1%	14.812
U/N Reef (21-094)	51.5	51.5	100%	0%	0%	0%	66.7%	11.727
Batt Reef (16-029)	830.2	58.9	7.1%	9.9%	23.4%	59.6%	76.2%	16.128
Bowden Reef (19-019)	147.6	101.1	68.5%	8%	20.8%	2.7%	76.2%	16.151
Lizard Island Reef (Lagoon) (14-116d)	67.7	19.0	28.1%	28%	35.7%	8.3%	76.7%	26.881
Judith Wright Reef (18-101)	144.4	56.6	39.2%	19%	25.2%	16.6%	79.2%	131.478
U/N Reef (21-092)	9.0	9.0	100%	0%	0%	0%	79.2%	41.938
U/N Reef (21-460)	59.9	59.9	100%	0%	0%	0%	8.3%	1.000
U/N Reef (21-466)	120.2	120.2	100%	0%	0%	0%	8.3%	0.938
U/N Reef (21-250a)	33.3	16.4	49.4%	36.6%	14%	0%	83.3%	15.536
Bushy-Redbill Reef (20-310)	35.4	0.0	0%	0%	0%	100%	87.5%	54.667
Banfield Reef (North) (18-105a)	531.6	37.7	7.1%	15%	29.6%	48.3%	91.7%	66.485
Davies Reef (18-096)	177.7	102.0	57.4%	27.4%	15.2%	0%	95.2%	71.802
(Little) Broadhurst Reef (18-106)	172.3	71.7	41.6%	29.7%	21.4%	7.4%	No Data	NA
Adelaide Reef (17-042)	13.3	13.3	100%	0%	0%	0%	No Data	NA
Arlington Reef (16-064)	48.5	32.4	66.8%	23.9%	9.3%	0%	No Data	NA
Backnumbers Reef (18-069a)	77.7	35.0	45.1%	54.9%	0%	0%	No Data	NA
Banfield Reef (South) (18-105b)	35.2	3.5	9.9%	41.5%	26.8%	21.8%	No Data	NA
Barnett Patches (18-019)	149.2	142.8	95.8%	4.2%	0%	0%	No Data	NA
Black Island Reef (20-017)	34.9	34.9	100%	0%	0%	0%	No Data	NA
Block Reef (19-127)	13.4	9.7	72.6%	27.4%	0%	0%	No Data	NA

Reef Name	Tot Dive Hrs	Wasted Dive Hrs	% 0-0.02	% 0.02-0.04	% 0.04-0.08	% >0.08	eDNA % Pos	eDNA Mean Conc
Boult Reef (23-079)	159.5	70.3	44.1%	16.6%	11.6%	27.7%	No Data	NA
Briggs Reef (16-074)	320.1	61.5	19.2%	17.9%	22.7%	40.2%	No Data	NA
Britomart Reef (18-024)	57.7	36.5	63.3%	36.7%	0%	0%	No Data	NA
Centipede Reef (18-088)	99.9	99.9	100%	0%	0%	0%	No Data	NA
Chinaman Reef (16-024)	106.0	50.6	47.8%	30.9%	21.3%	0%	No Data	NA
Circular Quay Reef (19-134)	74.2	24.1	32.4%	30.7%	26.7%	10.1%	No Data	NA
Darley Reef (19-043)	212.9	168.6	79.2%	8.1%	9.1%	3.7%	No Data	NA
East Cay Reef (21-305)	178.3	43.0	24.1%	23.2%	40.5%	12.3%	No Data	NA
Emily Reef (15-082)	76.3	0.0	0%	24.7%	50%	25.3%	No Data	NA
Emperor Reef (21-538)	166.1	123.3	74.2%	3.2%	11.5%	11.2%	No Data	NA
Evening Reef (15-095)	292.3	48.4	16.6%	31.7%	40.5%	11.2%	No Data	NA
Eye Reef (14-134)	135.1	0.0	0%	11.1%	37.5%	51.4%	No Data	NA
Fairfax Islands Reef (23-081)	22.8	14.6	63.7%	36.3%	0%	0%	No Data	NA
Fitzroy Island Reef (No 1) (16-054a)	7.2	7.2	100%	0%	0%	0%	No Data	NA
Fitzroy Reef (23-077)	333.0	143.6	43.1%	33%	23.5%	0.4%	No Data	NA
Fore And Aft Reef (18-043)	46.7	13.3	28.6%	42.9%	28.6%	0%	No Data	NA
Fork Reef (18-083)	43.0	35.8	83.3%	0%	16.7%	0%	No Data	NA
Forrester Reef (15-009)	281.1	124.3	44.2%	18.3%	15.9%	21.5%	No Data	NA
Grub Reef (18-077)	13.3	13.3	100%	0%	0%	0%	No Data	NA
Gull Reef (15-016)	14.6	6.0	41.1%	0%	58.9%	0%	No Data	NA
Hall-Thompson Reef (17-037)	25.3	13.3	52.8%	0%	47.2%	0%	No Data	NA

Reef Name	Tot Dive Hrs	Wasted Dive Hrs	% 0- 0.02	% 0.02- 0.04	% 0.04- 0.08	% >0.08	eDNA % Pos	eDNA Mean Conc
Hardy Reef (19-135)	21.6	18.4	85.5%	14.5%	0%	0%	No Data	NA
Harrier Reef (15-025)	42.0	42.0	100%	0%	0%	0%	No Data	NA
Hastings Reef (16-057)	33.9	33.9	100%	0%	0%	0%	No Data	NA
Hayman Island Reef (20-014)	23.3	19.3	82.9%	17.1%	0%	0%	No Data	NA
Helix Reef (18-076)	37.2	0.0	0%	0%	77.6%	22.4%	No Data	NA
Helsdon Reef (14-135)	341.8	36.5	10.7%	10.9%	41.4%	37%	No Data	NA
Heron Reef (23-052a)	360.4	272.9	75.7%	19.1%	2%	3.2%	No Data	NA
Holbourne Island Reef (No 1) (19-103a)	6.1	0.0	0%	100%	0%	0%	No Data	NA
Hook Island Reef (No 10) (20-028i)	28.7	28.7	100%	0%	0%	0%	No Data	NA
Hook Island Reef (No 6) (20-028f)	63.1	52.5	83.3%	10.4%	6.3%	0%	No Data	NA
Hook Island Reef (No 7) (20-028g)	27.4	25.7	93.9%	6.1%	0%	0%	No Data	NA
Hook Island Reef (No 8) (20-028j)	8.0	8.0	100%	0%	0%	0%	No Data	NA
Hook Island Reef (No 9) (20-028h)	29.1	29.1	100%	0%	0%	0%	No Data	NA
Hopkinson Reef (18-073)	312.0	161.3	51.7%	27.6%	9.9%	10.9%	No Data	NA
Hoskyn Islands Reef (23-080)	9.7	0.0	0%	100%	0%	0%	No Data	NA
Irene Reef (15-084)	1206.8	50.8	4.2%	14.7%	30.9%	50.2%	No Data	NA
Isobel Bennett Reef (21-505)	3.1	3.1	100%	0%	0%	0%	No Data	NA
Kelso Reef (18-030)	25.1	9.6	38.2%	61.8%	0%	0%	No Data	NA
Lady Elliot Island Reef (24-008)	57.1	38.0	66.5%	18.2%	15.3%	0%	No Data	NA
Lady Musgrave Reef (23-082a)	371.8	135.1	36.3%	22.9%	21.4%	19.3%	No Data	NA
Lamont Reef (23-076)	123.4	12.1	9.8%	53.2%	37%	0%	No Data	NA

Reef Name	Tot Dive Hrs	Wasted Dive Hrs	% 0-0.02	% 0.02-0.04	% 0.04-0.08	% >0.08	eDNA % Pos	eDNA Mean Conc
Langford-Bird Reef (20-019)	40.8	40.8	100%	0%	0%	0%	No Data	NA
Lark Reef (15-033)	15.0	0.0	0%	0%	100%	0%	No Data	NA
Linnet Reef (14-126)	66.2	22.5	34%	33.5%	32.5%	0%	No Data	NA
Little Fitzroy Island Reef (16-055)	31.2	0.0	0%	54%	46%	0%	No Data	NA
Little Kelso Reef (18-031)	120.0	26.7	22.2%	55.6%	22.2%	0%	No Data	NA
Llewellyn Reef (23-078)	410.1	138.9	33.9%	21.9%	32.9%	11.4%	No Data	NA
Low Islands Reef (16-028)	130.0	130.0	100%	0%	0%	0%	No Data	NA
Lynchs Reef (18-091)	87.5	83.5	95.4%	4.6%	0%	0%	No Data	NA
Mackay Reef (16-015)	210.4	69.0	32.8%	32%	17.9%	17.2%	No Data	NA
Mackay Reefs (15-024)	239.6	22.4	9.4%	8.5%	25%	57.2%	No Data	NA
Martin Reef (14-123)	108.7	72.2	66.4%	20.4%	13.2%	0%	No Data	NA
McSweeney Reef (11-016)	63.5	32.6	51.3%	44%	0%	4.7%	No Data	NA
Michaelmas Reef (16-060)	21.5	7.3	34.2%	0%	31.1%	34.8%	No Data	NA
Mid Reef (19-029)	310.6	56.9	18.3%	25%	31.3%	25.4%	No Data	NA
Morning Reef (15-098)	8.1	8.1	100%	0%	0%	0%	No Data	NA
One Tree Island Reef (23-055a)	462.1	88.2	19.1%	16.6%	32.3%	32%	No Data	NA
Pixie Reef (16-040)	105.0	15.1	14.4%	30.4%	7.9%	47.3%	No Data	NA
Potter Reef (No 1) (17-059a)	40.0	33.3	83.3%	16.7%	0%	0%	No Data	NA
Recreation Reef (21-501)	54.5	38.1	69.8%	21%	9.2%	0%	No Data	NA
Reg Ward Reef (18-017)	44.4	44.4	100%	0%	0%	0%	No Data	NA
Ribbon No 3 Reef (15-050)	19.0	19.0	100%	0%	0%	0%	No Data	NA

Reef Name	Tot Dive Hrs	Wasted Dive Hrs	% 0- 0.02	% 0.02- 0.04	% 0.04- 0.08	% >0.08	eDNA % Pos	eDNA Mean Conc
Round-Russell Reef (17-013)	10.0	5.0	50%	50%	0%	0%	No Data	NA
Shell Reef (19-028)	41.4	41.4	100%	0%	0%	0%	No Data	NA
South Direction Reef (North) (14-147a)	13.1	6.9	52.9%	47.1%	0%	0%	No Data	NA
Spitfire Reefs (East) (16-012b)	234.7	45.3	19.3%	16.2%	41.1%	23.4%	No Data	NA
St Crispin Reef (16-019)	92.4	92.4	100%	0%	0%	0%	No Data	NA
Startle Reefs (15-028)	302.9	19.2	6.3%	10.9%	3.7%	79.1%	No Data	NA
Stucco Reef (19-141)	40.8	30.3	74.3%	7.4%	18.3%	0%	No Data	NA
Swinger Reef (15-030)	62.4	2.5	4%	4%	28.6%	63.4%	No Data	NA
Sykes Reef (23-054)	76.4	42.3	55.4%	35.5%	9.2%	0%	No Data	NA
U/N Reef (10-372)	20.3	5.0	24.6%	75.4%	0%	0%	No Data	NA
U/N Reef (10-381)	18.2	4.0	22%	54.1%	23.9%	0%	No Data	NA
U/N Reef (10-384)	580.0	210.7	36.3%	31.8%	23.9%	8%	No Data	NA
U/N Reef (10-385)	37.3	26.8	71.7%	22.3%	6%	0%	No Data	NA
U/N Reef (10-386)	181.4	45.3	25%	45.1%	27.4%	2.5%	No Data	NA
U/N Reef (10-387)	12.0	12.0	100%	0%	0%	0%	No Data	NA
U/N Reef (10-400)	158.2	150.2	94.9%	5.1%	0%	0%	No Data	NA
U/N Reef (10-403)	21.7	21.7	100%	0%	0%	0%	No Data	NA
U/N Reef (10-414)	83.6	82.1	98.2%	1.8%	0%	0%	No Data	NA
U/N Reef (11-049)	114.9	23.5	20.4%	11.3%	35%	33.2%	No Data	NA
U/N Reef (14-133)	192.9	25.8	13.4%	24.8%	50.1%	11.8%	No Data	NA
U/N Reef (14-145)	14.7	10.9	74.2%	0%	25.8%	0%	No Data	NA

Reef Name	Tot Dive Hrs	Wasted Dive Hrs	% 0- 0.02	% 0.02- 0.04	% 0.04- 0.08	% >0.08	eDNA % Pos	eDNA Mean Conc
U/N Reef (14-149)	56.4	17.1	30.4%	35.6%	11.2%	22.8%	No Data	NA
U/N Reef (14-163)	92.5	0.0	0%	3.2%	8.4%	88.4%	No Data	NA
U/N Reef (15-023)	75.5	40.4	53.6%	0%	8.2%	38.2%	No Data	NA
U/N Reef (15-044)	49.4	0.0	0%	26.5%	40.3%	33.2%	No Data	NA
U/N Reef (15-045)	11.6	0.0	0%	43.8%	56.2%	0%	No Data	NA
U/N Reef (15-047)	147.8	10.6	7.2%	37.8%	46.4%	8.6%	No Data	NA
U/N Reef (15-049)	30.2	6.2	20.3%	50.8%	28.9%	0%	No Data	NA
U/N Reef (15-071c)	390.7	20.5	5.2%	21.5%	31.6%	41.7%	No Data	NA
U/N Reef (15-072)	14.3	10.6	73.9%	26.1%	0%	0%	No Data	NA
U/N Reef (16-016)	178.9	62.7	35%	9.1%	39.6%	16.3%	No Data	NA
U/N Reef (16-054f)	23.9	6.2	25.8%	24.4%	49.8%	0%	No Data	NA
U/N Reef (21-056b)	58.4	34.4	58.9%	16.2%	16.3%	8.6%	No Data	NA
U/N Reef (21-187)	43.6	16.2	37.2%	59.5%	3.2%	0%	No Data	NA
U/N Reef (21-188)	75.1	56.3	75%	11.5%	13.4%	0%	No Data	NA
U/N Reef (21-245)	67.2	46.4	69%	12.6%	12.1%	6.3%	No Data	NA
U/N Reef (21-441)	24.7	24.7	100%	0%	0%	0%	No Data	NA
U/N Reef (21-454)	21.6	21.6	100%	0%	0%	0%	No Data	NA
U/N Reef (21-465)	106.6	103.0	96.7%	0%	3.3%	0%	No Data	NA
U/N Reef (21-499)	23.9	16.3	68.2%	31.8%	0%	0%	No Data	NA
U/N Reef (21-500)	37.5	28.0	74.6%	25.4%	0%	0%	No Data	NA
U/N Reef (21-509)	21.0	21.0	100%	0%	0%	0%	No Data	NA

Reef Name	Tot Dive Hrs	Wasted Dive Hrs	% 0-0.02	% 0.02-0.04	% 0.04-0.08	% >0.08	eDNA % Pos	eDNA Mean Conc
U/N Reef (21-539)	23.9	4.3	18.2%	0%	12.9%	68.9%	No Data	NA
U/N Reef (21-550)	155.9	75.4	48.4%	5.7%	24.3%	21.6%	No Data	NA
U/N Reef (21-551)	311.6	92.1	29.5%	27%	20.4%	23.1%	No Data	NA
U/N Reef (21-553)	18.4	18.4	100%	0%	0%	0%	No Data	NA
U/N Reef (21-554)	65.0	65.0	100%	0%	0%	0%	No Data	NA
U/N Reef (22-088a)	261.1	65.9	25.2%	24.7%	22.4%	27.7%	No Data	NA
Wheeler Reef (18-095)	265.7	80.1	30.2%	14%	21.3%	34.6%	No Data	NA
Wilson Island Reef (23-050)	11.9	11.9	100%	0%	0%	0%	No Data	NA

Spatial Output: Risk Classification Map

To fully visualize how these operational recommendations fall out across the GBR, this interactive map projects every historical validation event natively against the field landscape. Because each Reef may be tested in multiple different years, you can natively toggle the analytical projection overlay by **Year** using the layers control widget on the right.

You can cross-reference our operational “Approach A” (Field Data Only logic) directly against “Approach B” (Integrated eDNA framework) classifications natively over the raw historical sampling footprints.

► Code

